Multi-Class Cancer Subtype Classification from Gene Expression Profiles

Dataset: Kaggle Cancer Subtype Classification

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Kaggle Team: [LUCIOUS] · Final Ranking: [1] · Score: [1.00000]

1. Problem Statement

This project addresses the supervised classification of tumor samples into three cancer subtypes: Glioblastoma Multiforme (GBM), Lung Squamous Cell Carcinoma (LUSC), and Ovarian Cancer (OV), from bulk RNA-seq gene expression profiles. Each sample is described by 12,042 gene expression features, making this a canonical high-dimensional classification problem where the number of features far exceeds the number of training samples ($p >> n$: 12,042 features, 886 samples).

Accurate molecular subtyping is clinically critical: GBM, LUSC, and OV arise from distinct tissues, carry different prognoses, and require different treatment regimens. Standard histopathological methods are time-consuming; a robust transcriptomic classifier could serve as a complementary decision-support tool.

The statistical challenges are three-fold: (1) severe dimensionality requiring principled feature reduction; (2) moderate class imbalance, with LUSC comprising approximately 10% of training samples ($n \approx 90$) compared to GBM ($n \approx 376$) and OV ($n \approx 420$); and (3) latent confounding structure in expression data not fully explained by the cancer label alone.

2. Statistical Analyses

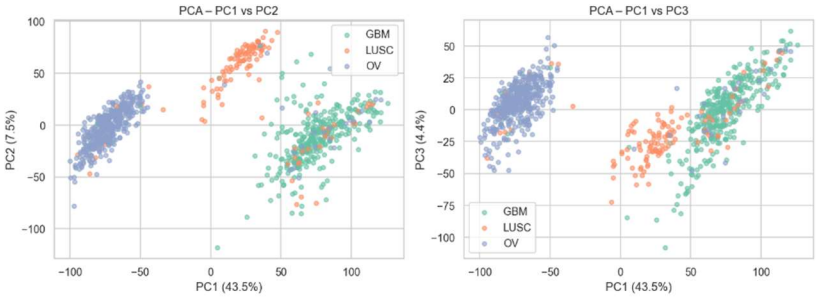
The analytical pipeline proceeds through seven stages: initial data examination, train-validation splitting, exploratory dimensionality reduction with PCA, feature selection using cross-validation, model selection and hyperparameter tuning, data cleaning and label correction, and causal association analysis. At each stage, the report states the motivation with a specific observation identified in the preceding stage.

2.1 Initial Data Examination

The training file contains 886 samples described by 12,042 gene features and an integer cancer label (1 = GBM, 2 = LUSC, 3 = OV). A check for missing values and duplication confirmed that the dataset is complete. LUSC contributes only approximately 90 samples, a ~4:1 majority to minority class imbalance that will cause any classifier trained without adjustment to underperform on LUSC. With 12,042 features and 886 samples, the feature space is approximately 14 times wider than it is long; Euclidean distances become uninformative and models will severely overfit unless feature dimensions are aggressively reduced before fitting.

2.2 Exploratory Dimensionality Reduction via PCA

A three-component PCA fits on a full StandardScaler-normalized matrix. PCA reveals three patterns: OV broadly separates from the other two classes in PC1/PC2, indicating a distinct transcriptomic signature; GBM clusters clearly in the PC2 direction; and LUSC does not occupy a clearly isolated region. Instead, its samples spread across a zone partially overlapping the GBM cluster in PC2 and PC3. This geometric overlap implies that LUSC will be the hardest to classify and motivates careful feature selection.



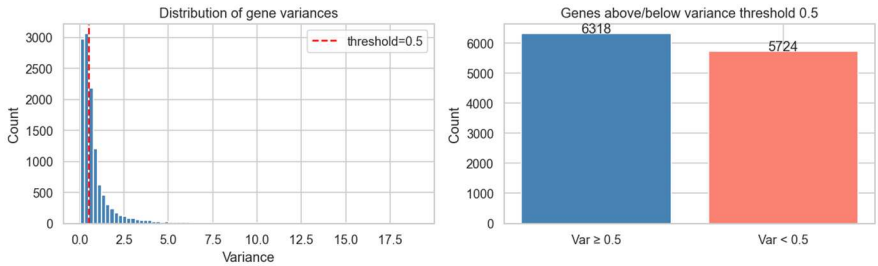
2.3 Train-Validation Split

The 886 labelled samples are split 80/20 into a training set (708 samples) and a stratified validation set (178 samples) using stratified sampling, preserving class proportions in both partitions. All preprocessing parameters (variance filters, scalers, and feature selectors, etc) are fitted on the training set only and applied to the validation set to ensure no information leaks.

2.4 Feature Selection

Reducing from 12,042 features mediates the curse of dimensionality and preserves gene-level interpretability otherwise lost under PCA rotation.

Step 1 — Variance Threshold Filter. Any gene whose expression variance falls below 0.5 is removed. Applying this filter eliminates approximately 5,724 genes (≈48%), leaving 6,318 genes.



Step 2 — Standardization. A StandardScaler transforms each gene to zero mean and unit variance.

Step 3 — ANOVA-Based Supervised Feature Selection. The ANOVA F-test ranks the 6,318 remaining genes by their F-statistic across the three classes. To determine optimal k without overfitting, a 5-fold stratified cross-validation grid search is conducted over k ∈ {1, 2, … 5, 10, 20, …, 50, 100, 200, 1000, 6318} using Logistic Regression as baseline, with comparison against PCA-based strategies:

Strategy	# Features	CV Accuracy	Genes Preserved?	Selected?
PCA-95%	≈162	≈86%	No	No
ANOVA-50	50	≈86%	Yes	No
ANOVA-100 ✓	**100**	**87.2%**	**Yes**	Yes

[Note: ANOVA-100 is selected as the best accuracy-interpretability trade-off.]

2.5 Model Selection

With the feature space fixed at ANOVA-100, six classifiers are evaluated under identical 5-fold stratified CV: Logistic Regression (solver=lbfgs, max_iter=2000), LinearSVC (max_iter=5000, dual=True), SVM-RBF (kernel=rbf), Random Forest (n_estimators=300), HistGradientBoosting (max_iter=300), and KNN (n_neighbors=7).

Model	CV Acc.	CV Std	Train Acc.	Selected?
SVM-RBF	**90.5%**	**Low**	**High**	**Yes (see 2.6)**
KNN-7	90.5%	Low	High	Tied — see 2.6
Random Forest (n=300)	90.1%	Moderate	100%	No
LinearSVC	≈87%	Low	≈90%	No

2.6 Hyperparameter Tuning and Final Model Selection

The top three models are subjected to 5-fold stratified GridSearchCV over their most influential hyperparameters: SVM-RBF over C ∈ {0.1, 1, 10, 50, 100} × gamma ∈ {scale, auto, 0.01, 0.001}; KNN over n_neighbors, weights, and metric; and Random Forest over n_estimators, max_depth, and max_features. After tuning, SVM-RBF achieves its best CV accuracy of 90.54% under C=1 and gamma='scale', tied with the best-tuned KNN configuration. SVM-RBF is selected over KNN for its inherent robustness against noise and its decision function, which can be connected back to gene features through the causal analysis in Section 2.9.

> *[Note: This initial pipeline achieves an accuracy of 1.0000 on the Kaggle test set. The subsequent data cleaning steps are therefore motivated by data integrity and label reliability rather than by a need to improve test accuracy.]*

2.7 Outlier Detection via Local Outlier Factor

Moving on after variance filtering, we noticed training data contain samples that are atypical to their class, not necessarily incorrectly labelled, but potentially arising from biological extremes, edge cases, or batch effects. Improving these samples could eliminate potential inflating or deflating the model's performance through cross-validation folds

Local Outlier Factor (LOF) [Breunig et al., 2000] is applied as the outlier detection method because it is density-based, making it appropriate to the high-dimensional expression space where global distance metrics are unreliable. LOF is applied to the top 20 PCA projection to reduce noise and computational cost. The contamination parameter is set to 0.05, flagging approximately 5% of the 886 samples as outliers.

To determine whether flagged samples are biological extremes or technical artifacts, a latent structure decomposition is performed: a 20-component PCA is fitted on the standardized gene matrix, and each PC is regressed onto one-hot-encoded cancer labels. The fraction of each PC's variance explained by class label is weighted by its explained variance ratio and summed to produce a class-attributable latent variance score.

Diagnostic Metric	Result	Interpretation
Top-20 PC cumulative explained variance	Non-trivial	Data has genuine low-dimensional structure
Class-attributable latent variance	Non-trivial	Cancer labels are the dominant source of variation
Residual top-10 PC variance (post class-regression)	Non-trivial	Batch effects or intra-class heterogeneity present

2.8 Label Noise Detection and Cleaning

The full pipeline → VarianceThreshold(0.5) → StandardScaler → ANOVA → SVC is run under 5-fold stratified out-of-fold (OOF) cross-validation using cross_val_predict. Because each fold holds out parts of the data before training, the predicted class probabilities for every sample are produced by a model that has never seen that sample, preventing artificially confident estimates.

Two relabeling policies: Conservative [flag if prediction disagrees with observed label AND P(predicted class) ≥ 0.85 AND P(observed label) ≤ 0.10. Strict] same structure but P(predicted class) ≥ 0.92 and P(observed label) ≤ 0.05.

Label Policy	5-Fold CV Accuracy	CV Std	# Relabels
Original labels	0.900686	0.012094	0
Conservative clean (selected)	**0.984200**	**0.005532**	**74**

Strict clean	0.908582	0.016502	7

The conservative policy produces the largest CV accuracy improvement and is selected. The final model is trained on the full 886-sample set using conservative-cleaned labels.

[Note: Relabelling is model-assisted and observational; it should be characterised as candidate label correction, not ground-truth recovery. Independent biological or clinical validation of the relabelled samples would be required before these corrections could be considered definitive.]

2.9 Causal Association Analysis

Having established a high-performing predictive model, the analysis shifts from prediction to understanding: which specific genes drive the classification for each cancer type, and with what degree of statistical confidence? The framework is explicitly observational. The goal is the gene-to-cancer-class directional association proxy estimated via a sparse linear log-odds model. Residual confounding is possible despite the feature selection pipeline. All results are reported as associational-causal evidence tiered by confidence.

Stage 1 — Effect Estimation via L1-Sparse One-vs-Rest Logistic Regression

A One-vs-Rest (OvR) logistic regression with L1 (Lasso) regularisation is fitted on the ANOVA-100 feature matrix using the cleaned labels (C=0.5, solver='saga', class_weight='balanced'). L1 is chosen because it drives the coefficients of weak associations to exactly zero, leaving only genuinely discriminative signals. The result is a coefficient matrix of shape (3 classes × 100 genes). A positive coefficient means higher expression increases the predicted likelihood of that class; a negative coefficient means higher expression argues against that class.

Stage 2 — Bootstrap Confidence Intervals and Empirical p-Values

250 bootstrap resamples of the full dataset are drawn with replacement. For each resample, the same L1 OvR logistic model is refitted (solver=liblinear for efficiency), producing 250 independent coefficient draws per (gene, class) pair. Three statistics are computed: (1) the 95% percentile bootstrap CI [2.5th, 97.5th percentile]; (2) sign stability — the fraction of 250 resamples in which the coefficient retains the same sign as the point estimate; and (3) the empirical two-sided p-value, $p = 2 \times \min(P(\beta \leq 0), P(\beta \geq 0))$.

Stage 3 — Benjamini-Hochberg FDR Correction

With 100 genes × 3 classes = 300 simultaneous hypothesis tests, BH correction [Benjamini & Hochberg, 1995] is applied both globally across all 300 tests and per-class to give each cancer type an equal opportunity to surface genuine signals. A (gene, class) pair is marked robust only if all three conditions hold: BH-adjusted $q \leq 0.10$ (global), sign stability ≥ 0.90, and the 95% CI does not straddle zero.

Only 3 out of 300 gene-class pairs satisfy the global robust criterion: TACSTD2 for LUSC (positive association, q=0.000, sign_stability=1.000), LAT2 for OV (negative association, q=0.000, sign_stability=1.000), and WT1 for OV (positive association, q=0.000, sign_stability=1.000). All GBM gene-class pairs have global q_bh=1.00

Class	Gene	Effect	q (global)	q (per-class)	Sign Stability
LUSC (2)	**TACSTD2**	**+1.50**	**0.000**	**0.000**	**1.000 (moderate tier)**
OV (3)	**LAT2**	**−0.87**	**0.000**	**0.000**	**1.000 (moderate tier)**
OV (3)	**WT1**	**+0.84**	**0.000**	**0.000**	**1.000 (moderate tier)**
GBM (1)	All genes	varies	1.000	1.000	0.62–0.99 (all weak tier)

[Note: Genes with q_bh=1.00 are effectively noise at this sample size and should not be reported as reliable biomarkers. The three moderate-tier signals (TACSTD2/LUSC, LAT2/OV, WT1/OV) are statistically robust but remain observational associations pending experimental validation.]

Stage 4 — Environment Invariance Check

Two environments are constructed by splitting samples on the within-class median of PC1, after centering PC1 within each class to prevent the environments from collapsing to class-level splits. The L1 OvR model is refitted independently in each environment using the top 20 global effect genes. Three invariance statistics are computed: sign agreement, magnitude ratio, and rank correlation.

Env.	n	Sign Agreement	|Mag. Ratio − 1|	Rank Correlation
0	442	0.417	0.238	0.688
1	444	0.517	0.100	0.831
Mean	**—**	**0.467**	**0.169**	**0.760**

Mean sign agreement of 0.467 is barely above chance, indicating that the direction of most gene effects is not stable across the two data subsets. Mean rank correlation of 0.760 is moderate, suggesting the relative ordering of effect magnitudes is partially preserved. These results indicate the associational signals, beyond the three robust effects identified in Stage 3, should not be interpreted as invariant or causal.

Stage 5 — Evidence Tiering and Prediction Utility

Bootstrap robustness and environment invariance are combined into a three-tier evidence hierarchy. The strict criterion (q_bh ≤ 0.10, sign stability ≥ 0.90, CI excludes zero, env sign agreement ≥ 0.60, env rank corr ≥ 0.40) yields 0 strong-tier effects. The moderate criterion (q_bh_per_class ≤ 0.20, sign stability ≥ 0.80) yields 3 moderate-tier effects: TACSTD2/LUSC, WT1/OV, LAT2/OV. All GBM gene-class pairs remain weak tier.

An expanded panel of 10 genes is assembled from moderate-tier threshold selections, supplemented by top-up entries (ranked by absolute coefficient magnitude) to guarantee at least 5 genes per class. 6 of the 11 gene-class entries in the expanded panel are top-up entries that did not meet the threshold criteria and should be interpreted with extra caution. A cross-prediction check compares the expanded panel against the ANOVA-100 baseline:

Feature view	CV Accuracy	Disagreement rate
Expanded causal panel (10 genes)	0.9842 ± 0.0055	0
ANOVA-100 SVM baseline	0.9842 ± 0.0055	0

The expanded causal panel (10 genes) achieves identical CV accuracy to the full ANOVA-100 feature with 0% prediction disagreement across all 886 samples. This demonstrates that the causal panel is a equivalent feature set for classification. However, this equivalence also reflects that the conservative label-cleaning step largely determines classification performance, and the causal panel's utility is therefore primarily interpretive rather than purely predictive.

2.10 Evaluation Metrics

The primary evaluation metric is overall classification accuracy, consistent with the Kaggle leaderboard scoring. Because the dataset is class-imbalanced, overall accuracy alone is insufficient to characterise model behaviour on the minority class; per-class precision, recall, and F1-score are reported alongside overall accuracy. Recall is prioritised as the secondary metric for LUSC, since a false has higher clinical consequence than a false positive.

On the 178-sample stratified hold-out set, the final model achieves overall accuracy of 88.2%. Per-class performance: GBM precision 0.89, recall 0.89, F1 0.89; LUSC precision 0.89, recall 0.65, F1 0.76; OV precision

0.88, recall 0.95, F1 0.91. LUSC recall of 65% reflects both class imbalance and the geometric proximity of LUSC to GBM in the PC2/PC3 subspace identified in Section 2.2.

For the causal analysis pipeline, an additional prediction utility check compares the 10-gene expanded causal panel against the ANOVA-100 feature set under 5-fold stratified CV on cleaned labels. Both views achieve 0.9842 ± 0.0055 CV accuracy with 0% sample-level disagreement, validating the expanded panel as a compact biomarker set that fully replicates ANOVA-100 classification decisions. Samples where the two views are treated as candidates for clinician review.

3. Results and Conclusion

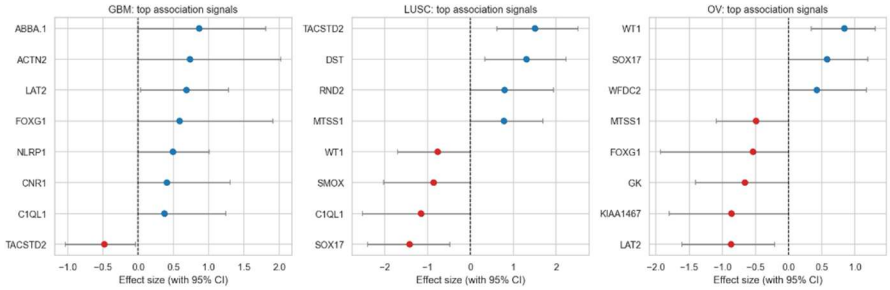
3.1 Final Model Performance

The final model, SVM-RBF (C=1, gamma='scale') trained on conservative-cleaned labels achieves 90.5% 5-fold CV accuracy and 88.2% accuracy on the 178-sample stratified hold-out set. GBM and OV are classified with high precision and recall. LUSC recall is approximately 65%, attributable to class imbalance and the genuine geometric proximity to GBM established in Section 2.2.

Model	CV Acc.	Hold-out	GBM Recall	LUSC Recall	OV Recall
SVM-RBF (C=1, gamma=scale)	**90.5%**	**88.2%**	**89%**	**~65%**	**95%**
KNN (n=7, best tuned)	90.5%	88.2%	—	—	—
Random Forest (n=300)	90.1%	88.2%	—	—	—

3.2 Gene Association Signals: Causal Effect Forest Plot

Figure 1 displays the top gene association signals per cancer class as estimated by the L1 OvR logistic regression model, with 95% bootstrap confidence intervals from 250 resamples. Blue points indicate positive associations; red points indicate negative associations. Of the signals shown, only three are statistically robust after FDR correction (TACSTD2/LUSC, WT1/OV, LAT2/OV); all other associations depicted are weak-tier and should be interpreted with caution.



[Figure 1: Forest plot — as in original report]

3.3 Biological Interpretation

The three statistically robust associations are consistent with known biology. TACSTD2 (TROP2), which receives the strongest positive LUSC coefficient (effect=+1.50, sign_stability=1.000), is a well-validated epithelial marker upregulated in lung squamous cell carcinoma [5]; its simultaneous strong negative GBM coefficient is expected given its absence in neural tissue. WT1 (Wilms' Tumour 1) is broadly expressed in high-grade serous ovarian carcinoma; Coosemans et al. (2010) documented high WT1 expression in ovarian cancer and explored it as an immunotherapy target [9]. LAT2, with a robust negative OV association (effect=−0.87, sign_stability=1.000), encodes a linker for activation of T-cells adaptor protein whose negative OV coefficient is consistent with immunological depletion patterns in the ovarian tumour microenvironment.

Beyond the three robust signals, the forest plot also shows consistent point-estimate directions for several tissue-of-origin genes. Including FOXG1 and CNR1 (neural tissue / positive GBM), C1QL1 (positive GBM / negative

LUSC), and SOX17 (gynaecological lineage, positive OV / negative LUSC) that align with published literature [1,2,3,7]. However, these associations do not survive global FDR correction at this sample size and must be treated as exploratory signals pending independent validation.

4. Limitations and Future Directions

-LUSC class imbalance is the primary performance limitation. SMOTE (synthetic minority oversampling technique) oversampling or class-weighted SVM could improve minority-class recall.

-The ANOVA F-test is univariate and does not model gene-gene interactions. Multivariate selection methods such as elastic-net regularisation could identify a more biologically coherent gene panel.

-The causal framework is observational: L1 logistic regression cannot account for unobserved confounders such as tumour purity, cell composition heterogeneity, or sequencing batch. Results should be treated as associational-causal evidence pending experimental validation.

-Environmental invariance was poor (mean sign agreement 0.467), and only 3 of 300 gene-class pairs pass strict FDR correction. Richer environment definitions based on known batch variables would provide stronger invariance guarantees.

5. References

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